

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer each and every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Draw a function that is continuous everywhere except for $x = -2, 1, \text{ \& } 5$. Use a different type of discontinuity at each of these values of x .

2. Determine if the following function is continuous at $x = -2$. $f(x) = (x+1)^2$.

3. Is the following piecewise function continuous at $x = -2$?

$$f(x) = \begin{cases} (x+1)^2, & x \neq -2 \\ 1, & x = -2 \end{cases}$$

4. Is the following piecewise function continuous at $x = -2$.

$$f(x) = \begin{cases} \frac{x^2 - x - 6}{x^2 + 5x + 6}, & x \leq 0 \\ \frac{\ln(x+3)}{e^{x+2}}, & x > 0 \end{cases}$$

5. Is the following function continuous at $x = -2$? Justify your answer.

$$f(x) = \begin{cases} \frac{x^2 - x - 6}{x^2 + 5x + 6}, & x \leq -2 \\ \frac{\ln(x+3)}{e^{x+2}}, & x > -2 \end{cases}$$

6. Where is the function $y = \frac{e^{\frac{1}{x-1}} + \sin(3x^2 + 2)}{\tan(x)}$ continuous?

7. For what value of the constant c is the function f continuous for all real numbers?

$$f(x) = \begin{cases} cx^2 + 5x & \text{if } x < 3 \\ x^3 - cx & \text{if } x \geq 3 \end{cases}$$

8. Consider the function $f(x) = \begin{cases} \frac{x^2 - 9}{x - 3} & x < 3 \\ ax - 1 & x \geq 3 \end{cases}$.

a) For what value of a is the function continuous at $x = 3$?

b) If $a = 5$, what type of discontinuity exists at $x = 3$? Justify your answer.

The following questions are related to the Intermediate Value Theorem (IVT)

9. Something is wrong with the following ‘proof’. What is it?

Let $f(x) = \frac{x}{|x|}$. When $x = 1$, $f(x) = 1$. When $x = -1$, $f(x) = -1$. There, by the IVT,

$f(c) = 0$ for some c between -1 and 1 .

10. Let C be the ellipse $x^2 + 4y^2 = 1$. How can one use the IVT to show that there is point P on C such that the area of the triangle determined by $(1,0)$, $(-1,0)$ and C is $\frac{\pi}{12}$. Hint: How does the area of the triangle change as one chooses different points on C ?

11. Show that there exists a rectangular box with edges of length x , $x+1$, and $x+2$ for some positive value x so that the volume of the box is equal to its surface area (assume the box has both a top and a bottom).